# 2018

## COMPUTER SCIENCE

**Paper : CSM-103**

**(Advances in Data Structure)**

**Full Marks : 70**

*The figures in the margin indicate full marks.*

*Candidates are required to give their answers in their own words as far as practicable.*

Answer *question nos.* 1, 2 and *any four* questions from the rest.

1. Answer *any five* questions from the following : 2×5

   (a) Define 'little-oh'. Justify the statement : 'In the o-notation, the function $f(n)$ becomes insignificant relative to $g(n)$ as $n$ approaches infinity'.

   (b) What do you mean by *mergeable heaps*? Give examples.

   (c) Write the constraints and the complexity of *RadixSort*.

   (d) State the necessity of introducing *double rotation*, instead of *single rotation*, in balancing the height of an AVL tree.

   (e) Define the *largest descendant functions* of a 2-3 tree. State their significance.

   (f) Why an unique binary tree can't be built from Preorder and Postorder traversal?

   (g) How many comparisons are needed to sort two individually sorted list of size $m$ and $n$ respectively?

   (h) What is Open Hashing?

2. Answer *any five* questions from the following :

   (a) Write down the necessity of incoming stack and in-stack priority while converting an infix notation to its postfix notation. Why is it necessary to compute the postfix notation? 3+1

   (b) Use necessary assumption(s), if any, and give an outline of an algorithm to convert a three-dimensional array to its two-dimensional equivalent. Also, cite a suitable example to support your algorithm. 2+2

   (c) Define tree edge, forward edge, back edge, and cross edge related to DFS on a graph. Show that an undirected graph does not contain any cross edge. 2+2

   (d) Define hashing. Describe at least three methods of building a hash function and relatively compare them. 1+3

   (e) What do you mean by amortized analysis of algorithms? How is it differentiated from worst-case analysis and average-case analysis? State the role of the credit balance function in computing the amortized cost of some algorithm. 1+2+1

(f) Consider the following sequences of a tree :

Pre-order : G, B, Q, A, C, K, F, P, D, E, R, H

In-order : Q, B, K, C, F, A, G, P, E, D, H, R

Construct the post order sequence. 4

(g) How Binary Search Tree differs from Binary max Heap? What is the worst case searching complexity in Binary search tree? Explain with example. 2+1+1

(h) Write the properties of B-tree. What is the searching complexity in B-tree? 2+2

(i) What is linear probing? What is clustering in a hash table? 2+2

3. (a) State a problem whose solution strategy might require stack as a data structure to solve it. Briefly outline the algorithm and show how stack can be utilized.

(b) Let $X$ be an $n \times n$ matrix, where $3 \leq n \leq 12$, such that the elements in each row are in non-decreasing order but the elements in each column are in non-increasing order. Draw a decision tree that finds the first three largest elements in $X$ and their suquences in non-increasing order in the *minimum* number of comparisons. (2+4)+4

4. (a) Briefly describe the similarities and dissimilarities of Binomial heaps and Fibonacci heaps, including their memory representation.

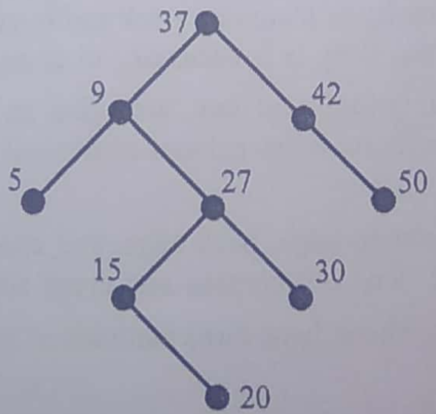
(b) Prove that there is at most one binomial tree in a binomial heap whose root has a given degree.

(c) Show how *SieveSort* can be used to determine the $k$-th smallest element of a list of $n$ distinct elements, where $k \leq n$. 4+3+3

5. (a) Define sparse matrix. Why is it so called? Relatively compare the static and dynamic representations of a sparse matrix to store its non-zero elements.

(b) Write a short note on threaded binary tree. (1+1+4)+4

6. (a) Define AVL tree. Why is it necessary to make a tree height balanced? Consider the following binary tree and balance it, if it is not.




Perform the following operations and at the end of each operation and balance the tree in case it has lost the property of an AVL tree. Insert 33; Insert 54; Delete 20.

(b) Construct an equivalent binary tree of the following forest that comprises $T_1$ and $T_2$. Clearly show the necessary steps of the construction.

$T_1$ : (A(B(E,F), C(G), D(H(K), I(L,M),J))), and

$T_2$ : (N(O(P))). (1+2+4)+3

7. Write the algorithm to convert an infix expression to postfix. Explain circular queue with an example. Write two disadvantages of array implementation of polynomial ADT. 5+3+2

8. Create a B-tree of order 5 with the elements 10, 40, 35, 15, 60, 25, 45, 70, 55, 30, 90, 50, 85, 99, 20, 65, 75, 95, 97, 80. Show the tree after deletion of the elements 55 and 60. 6+4

9. What is meant by comparison based sort? Apply Heap sort algorithm to sort the following list of elements 35, 13, 7, 15, 21, 6, 11, 12. Analyze the time complexity of Heap sort. 2+5+3